Panskura Banamali College (Autonomous)

B.Sc.(H) 6th Semester Examination 2022 (Online mode)

Subject : BCA **Paper: CC14P**

Paper Name : Theory of Automata and Compiler Design Lab

F.M. : 20 **Time : 2 Hours**

Instruction

[Make your answer script into PDF and MP3 file format and file name as like candidate_id-paper_code.pdf . For Example – candidate_id: 2020UG009967, paper_code : CC13P , then file name will be :: 2020UG009967- CC13P.pdf for question 'A' and 2020UG009967- CC13P.mp3 for question 'B']

A. **Answer any one question**: 1 x12 = 12

1. Lex program to find the number of constants.
2. Lex program to count the number of identifiers.
3. Lex program to count the number of comment lines.
4. Lex program to count the number of printf and scanf statements.
5. Lex Program to convert abc to ABC.
6. Lex program to find out total number of vowels, and consonants from the given input sting.
7. Lex program to check the validity of arithmetic statement.
8. Lex program to find simple and compound statements.

Marks distribution:

a. Problem Definition - 1 Marks
b. Algorithm/Flow Chart/ Pseudo-code– 5 Marks
c. Program listing /coding- 3 Marks
d. Output/Result with discussion- 3 Marks

B. Answer any four questions: 4 x2 = 8

1. Define compiler.
2. Define cross-compiler.
3. List various language Translators.
4. Is it necessary to translate a HLL program? Explain.
5. List out the phases of a compiler?
6. Which phase of the compiler is called an optional phase? Why?
7. Explain the working of a NFA?
8. When do you prefer to design an NFA to DFA?

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